

Fundamental Fluid Mechanics

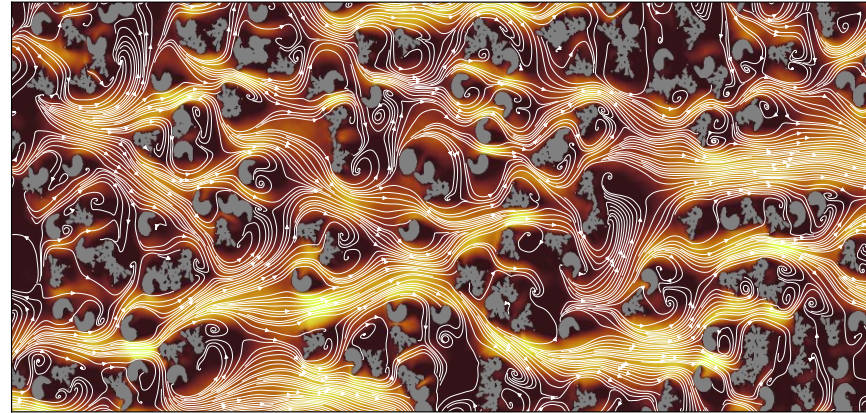
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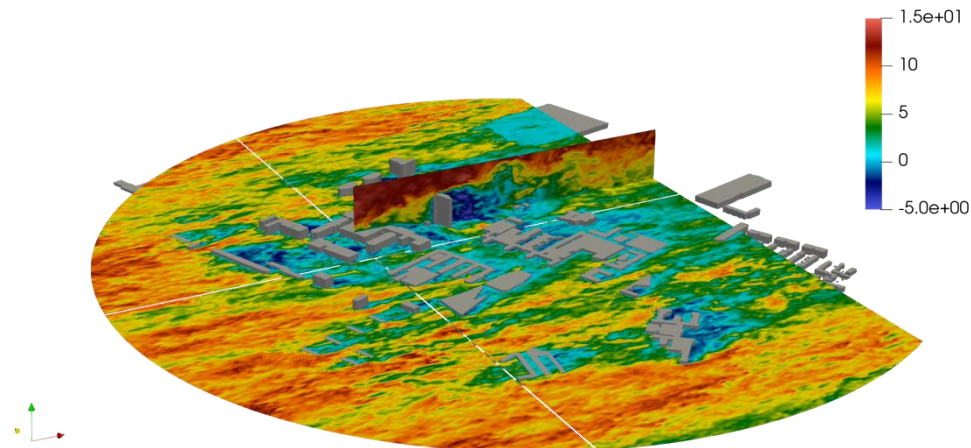
OpenFOAM Hackathon | Guimarães, Portugal 2026



Post-doctoral Researcher 3D Geoinformation Research Group
PhD in Environmental Fluid Mechanics, Stanford University

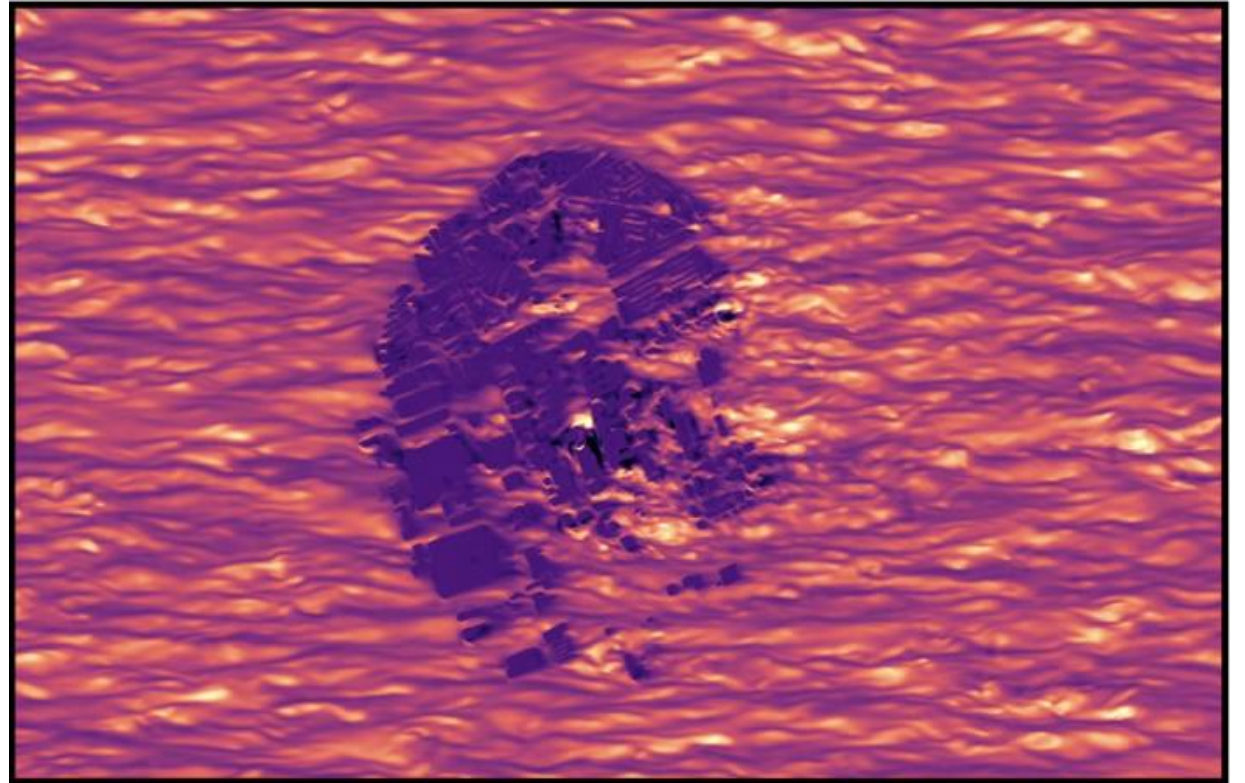


Associate Prof. 3D Geoinformation Research Group
PhD in Physics, 2017, von Karman Institute - Antwerp University



AGENDA

- ❑ Governing Equations
- ❑ Scales of Turbulent Motion
- ❑ Turbulence Modelling
- ❑ Wall modelling



Turbulent Flow Over TU Delft Campus

GOVERNING EQUATIONS



To understand fluid flow - Define what constitutes a "*fluid*"

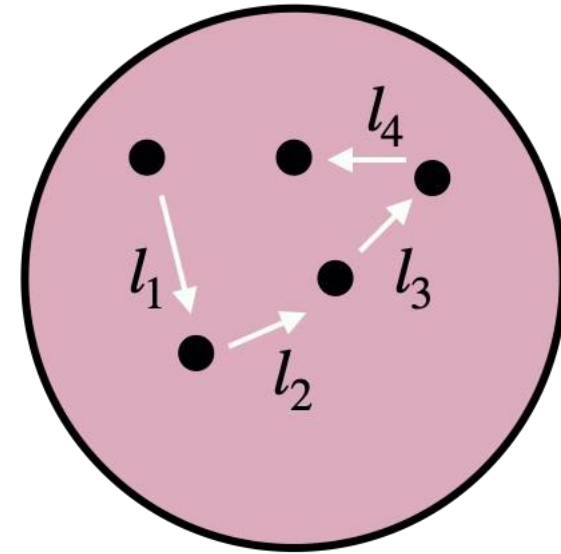
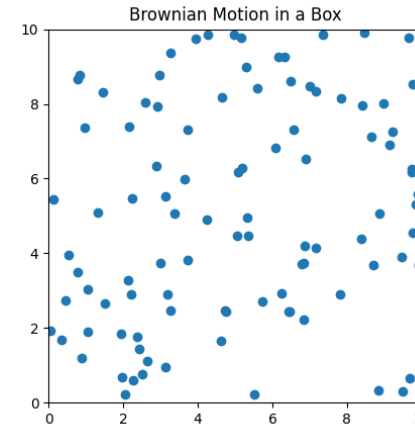
Fluid at the microscopic scale has "*chaotic*" and "*random*" motion

Smallest scales - Statistical Physics/Mechanics

We want to understand the macroscopic physics

Challenge: How do we develop this physics framework?

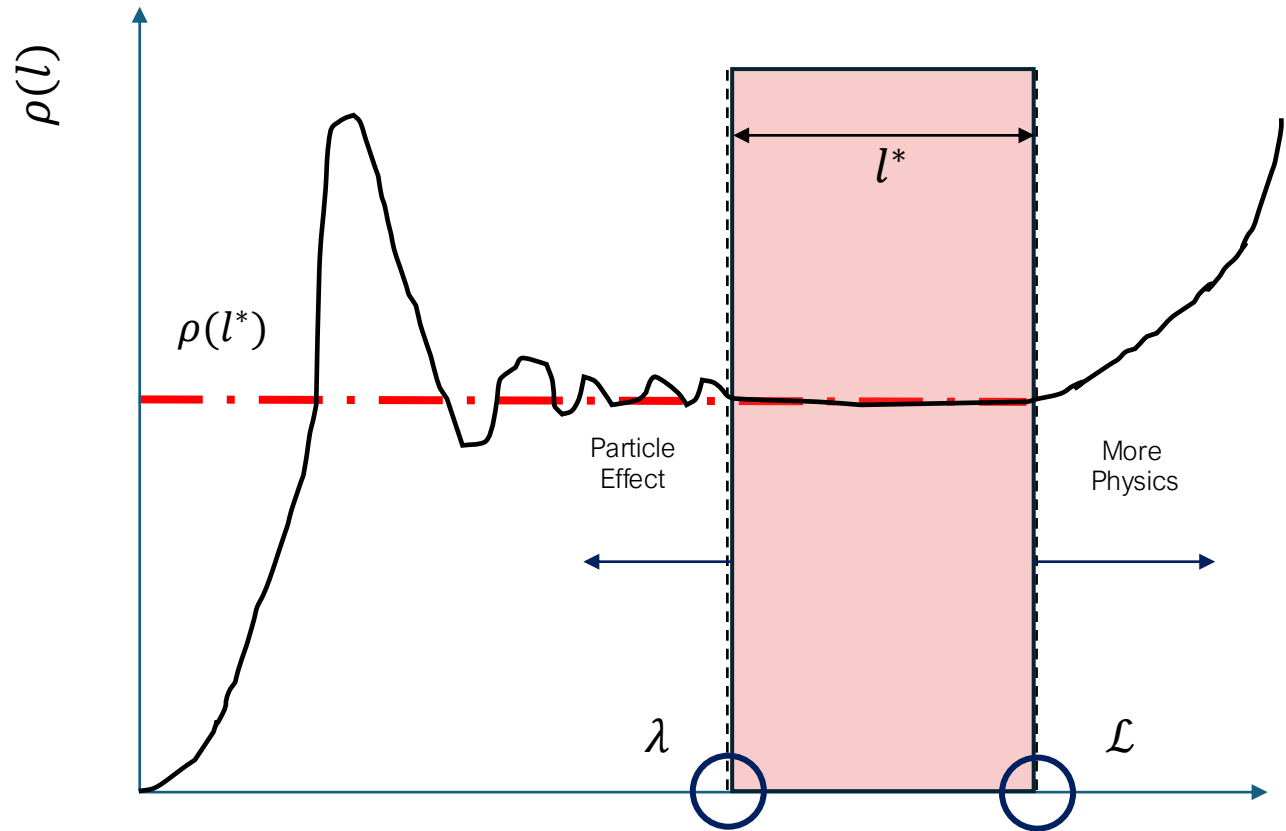
Solution: Continuum hypothesis (Not the Zermelo-Fraenkel set theory!)



GOVERNING EQUATIONS



λ – Mean Free Path $\mathcal{L}$ – Macroscopic Length Scale of Interest



$$\lambda = \frac{1}{N} \sum_{i=1}^{i=N} l_i$$

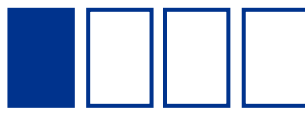
Knudsen number

$$\mathfrak{K}_n = \frac{\lambda}{\mathcal{L}} \ll 0(10^0)$$

Continuum works (usually) till $\mathfrak{K} = 0.01$

Now that we have averaged everything, we need to decide how to study macroscopic fluid flow in response to macroscopic external forces.

GOVERNING EQUATIONS



Averaging allows us to define FIELDS and the concept of a FLUID

What is a fluid

A fluid is a material that "flows"

Any applied stress leads to continuous deformation

Any applied stress results in a strain rate (thus, velocity)

Recommended Reading

Kundu, Cohen, Dowling 4th Edition

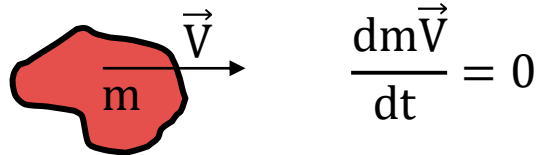
1.7 Fluid Statics

1.8 Classical Thermodynamics

1.9 Perfect Gas

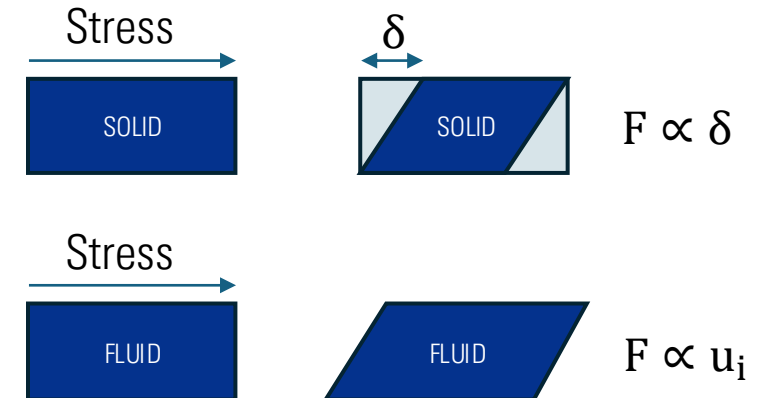
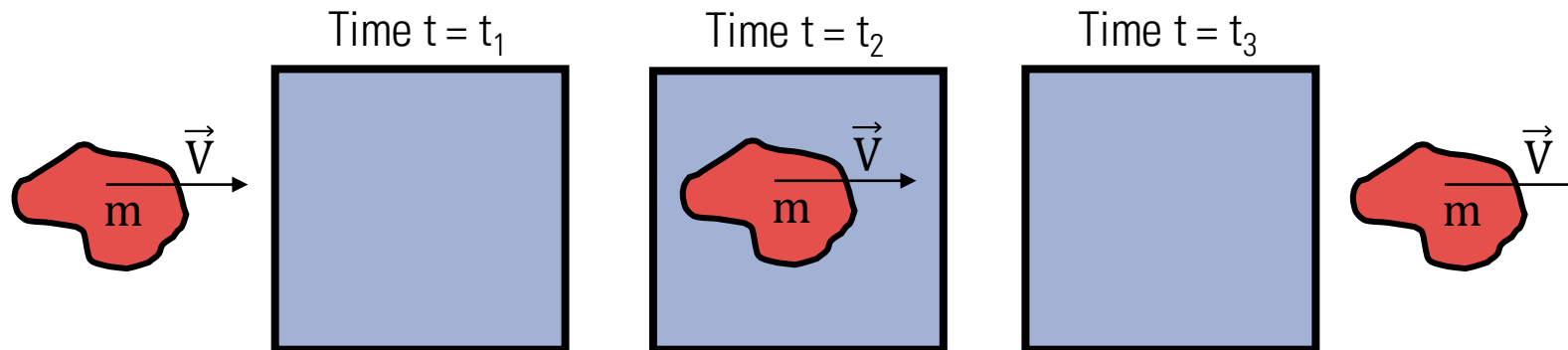
Thought Experiment

Newtons 2nd Law



$$\frac{dm\vec{V}}{dt} = 0$$

Eulerian Perspective



Until the Stress/Force is applied,
fluids flow!

GOVERNING EQUATIONS

Question: How does a field evolve along Lagrangian trajectories?

Solution: Write Lagrangian derivative that can be applied to Eulerian perspective.

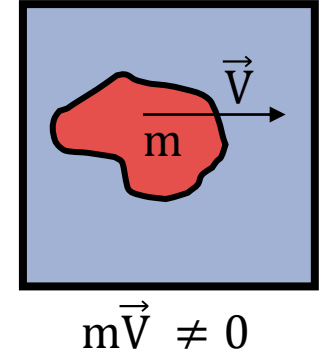
$$d\Phi = \left[\frac{\partial \Phi}{\partial t} dt + d\vec{X} \cdot \nabla_{\vec{X}} \Phi \right]_{\vec{X}=\vec{X}} \xrightarrow{d/dt} \frac{d\Phi}{dt} = \left[\frac{\partial \Phi}{\partial t} + \frac{d\vec{X}}{dt} \cdot \nabla_{\vec{X}} \Phi \right]_{\vec{X}=\vec{X}}$$

$\frac{\partial \Phi}{\partial x} \hat{i} + \frac{\partial \Phi}{\partial y} \hat{j} + \frac{\partial \Phi}{\partial z} \hat{k}$

$$\frac{d\Phi}{dt} = \left[\frac{\partial \Phi}{\partial t} + U(\vec{X}, t) \cdot \nabla_{\vec{X}} \Phi \right]_{\vec{X}=\vec{X}} \longrightarrow \boxed{\frac{d\Phi}{dt} = \left[\frac{\partial \Phi}{\partial t} + \vec{u}(\vec{x}, t) \cdot \nabla_{\vec{x}} \Phi \right]_{\vec{x}=\vec{x}}}$$

Hint: Read the Reynolds transport theorem for a more rigorous proof

WARNING: We used an illegal mathematical operation here to reach the conclusion!!!



$\frac{\partial \Phi}{\partial t}$ - Fixed position $\vec{x}$

$\frac{d\Phi}{dt}$ - Fixed fluid element $\vec{y}$

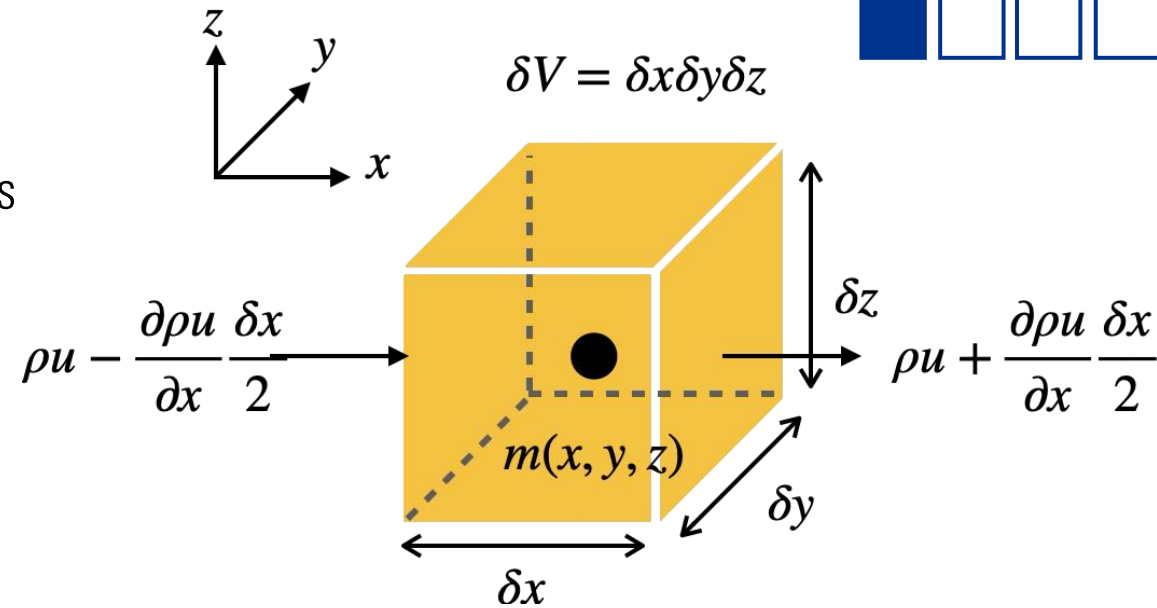
$$\Phi(t, \vec{y}) = \phi(X(t, \vec{y}), t)$$

What is X at time t for a particle which was at y at t_0

GOVERNING EQUATIONS

Lagrangian Principle: Newton's First Law - Total change in the mass must be zero

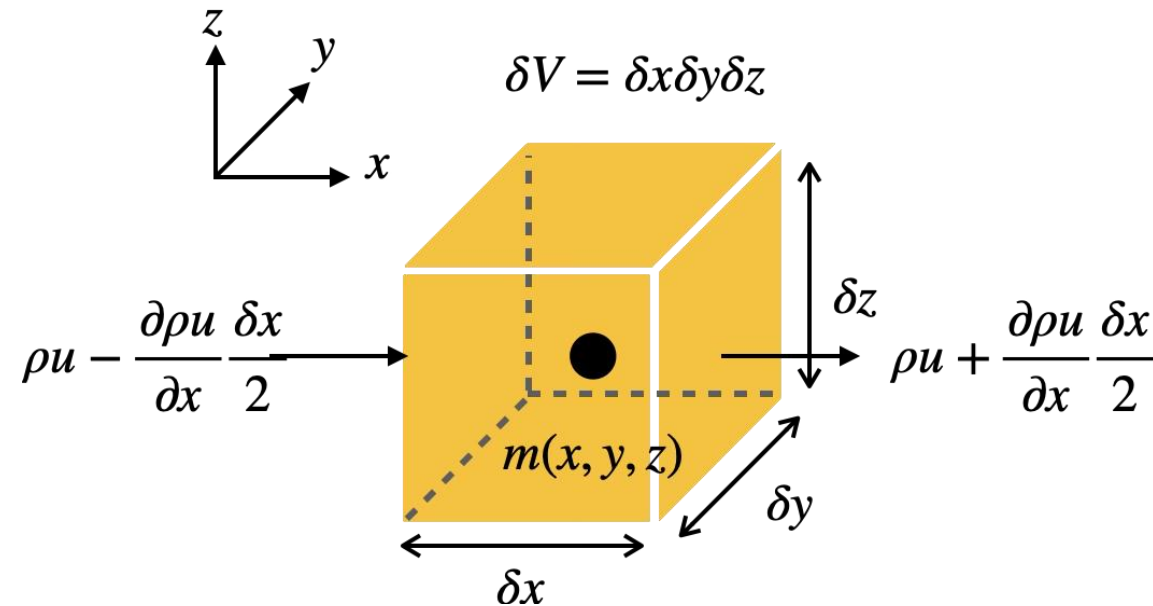
$$m [kg] = \rho \left[\frac{kg}{m^3} \right] \cdot V [m^3]$$



$$\begin{aligned} \frac{\partial \rho}{\partial t} \delta x \delta y \delta z &= (\rho u - \frac{\partial \rho u}{\partial x} \frac{\delta x}{2}) \delta y \delta z - (\rho u + \frac{\partial \rho u}{\partial x} \frac{\delta x}{2}) \delta y \delta z \\ &+ (\rho v - \frac{\partial \rho v}{\partial y} \frac{\delta y}{2}) \delta x \delta z - (\rho v + \frac{\partial \rho v}{\partial y} \frac{\delta y}{2}) \delta x \delta z \\ &+ (\rho w - \frac{\partial \rho w}{\partial z} \frac{\delta z}{2}) \delta x \delta y - (\rho w + \frac{\partial \rho w}{\partial z} \frac{\delta z}{2}) \delta x \delta y \end{aligned}$$

Conservation of Mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y} + \frac{\partial \rho w}{\partial z} = 0 \quad \longrightarrow \quad \frac{1}{\rho} \frac{d\rho}{dt} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$





$$\frac{1}{\rho} \frac{d\rho}{dt} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Special Fluid: Density invariance to space and time

THINK!

Requires a more “rigorous” proof than what we will do today.
Also called the Boussinesq Approximation - Further Reading Kundu & Cohen!

GOVERNING EQUATIONS



While we work in 3-dimensions, we will need some tools that will make writing Long equations a little bit simpler!

Scalar: Only needs a magnitude

Vector: Needs a magnitude and direction

Some Rules for 3D space

1. Summation Rule $\vec{x} \equiv x_i \hat{e}_i = x_i$ Repeating indices imply summation

$$\circ \delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33}$$

$$\circ \alpha_{iji} = (\alpha_{111} + \alpha_{212} + \alpha_{313})\hat{e}_1 + (\alpha_{121} + \alpha_{222} + \alpha_{323})\hat{e}_2 + (\alpha_{131} + \alpha_{232} + \alpha_{333})\hat{e}_3$$

2. Order agnostic notation - Unique indices

$$\circ \delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33}$$

$$\circ \alpha_i$$

$$\circ \gamma_{ij}$$

$$\circ \zeta_{ijk}$$

0th rank

1st rank

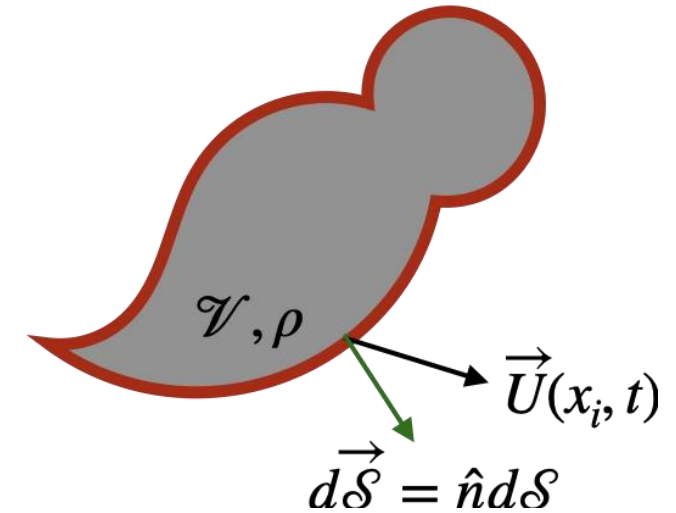
2nd rank

3rd rank

3. Tensor Multiplication

$$\circ a_{ij}d_{jkm} = f_{ikm} \quad \text{Similar to dot product/ inner product}$$

$$\circ b_{ij}f_{klm} = g_{ijklm} \quad \text{Tensor multiplication}$$

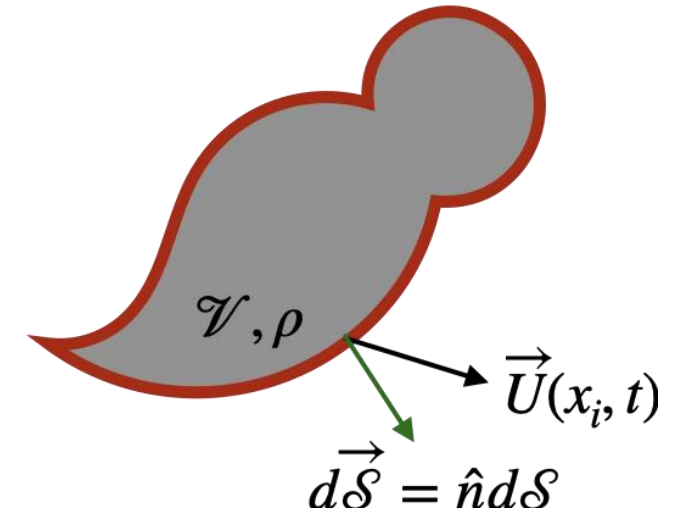


4. Gradient operator

$$\nabla \equiv \hat{e}_i \frac{\partial \langle \cdot \rangle}{\partial x_i} = \partial_i \langle \cdot \rangle$$

SUMMARY

- | | |
|--|-------------------------|
| 1. Addition | $n + n = n$ |
| 2. Multiplication | $n \times m = n + m$ |
| 3. Contraction | $n(n \geq 2) = n - 2$ |
| 4. Inner Product | $n \cdot m = n + m - 2$ |
| 5. Gradient
(Itself 1st
rank tensor) | $n, m = n + m$ |



We will see more tensor's but for now this is enough!

Recall - Leibniz-Reynolds Transport Theorem

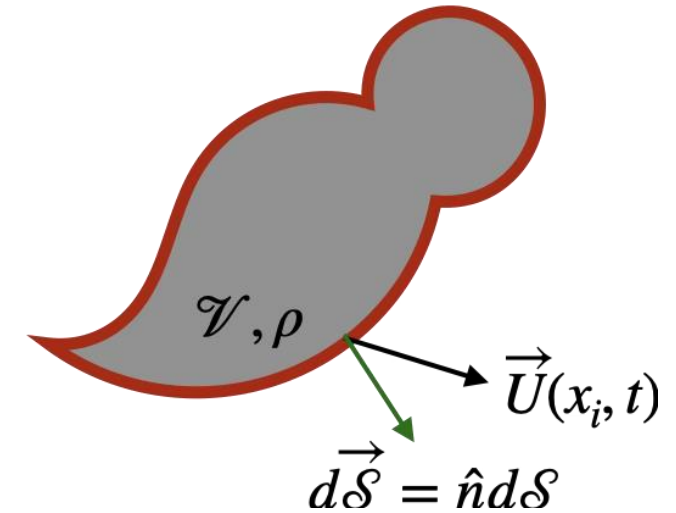
Formal - LRTT

$$\frac{d}{dt} \int_{\mathcal{V}(t)} \phi(x_i, t) d\mathcal{V} = \int_{\mathcal{V}(t)} \frac{\partial \phi(x_i, t)}{\partial t} d\mathcal{V} + \boxed{\int_{\mathcal{S}(t)} \phi(x_i, t) \vec{U}(x_i, t) \cdot d\vec{\mathcal{S}}}$$

Gauss-Divergence Theorem

$$\frac{d}{dt} \int_{\mathcal{V}(t)} \phi(x_i, t) d\mathcal{V} = \int_{\mathcal{V}(t)} \frac{\partial \phi(x_i, t)}{\partial t} + \nabla \cdot [\vec{U}(x_i, t) \phi(x_i, t)] d\mathcal{V}$$

Nothing special about the volume!



GOVERNING EQUATIONS



Lagrangian Principle: Newtons Second Law - Total change in the momentum must be zero

Consider an arbitrary volume: $V(t)$

$$\frac{d}{dt} \int_{V(t)} \rho \vec{U} dV \equiv_{RTT} \int_{V(t)} [\partial_t \rho u_i + \partial_j \rho u_j u_i] dV = \sum B + \sum S$$

$\int_{V(t)} \rho g_i dV$
 $\int_{V(t)} \partial_j \tau_{ij} dV$

Putting Everything Together

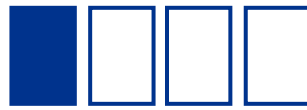
$$\partial_i u_i = 0$$

$$\partial_t u_i + \partial_j u_j u_i = g_i + \frac{1}{\rho} \partial_j \tau_{ij}$$

1. Continuum Hypothesis

2. Leibniz-Reynolds Transport Theorem

3. Incompressible Fluid

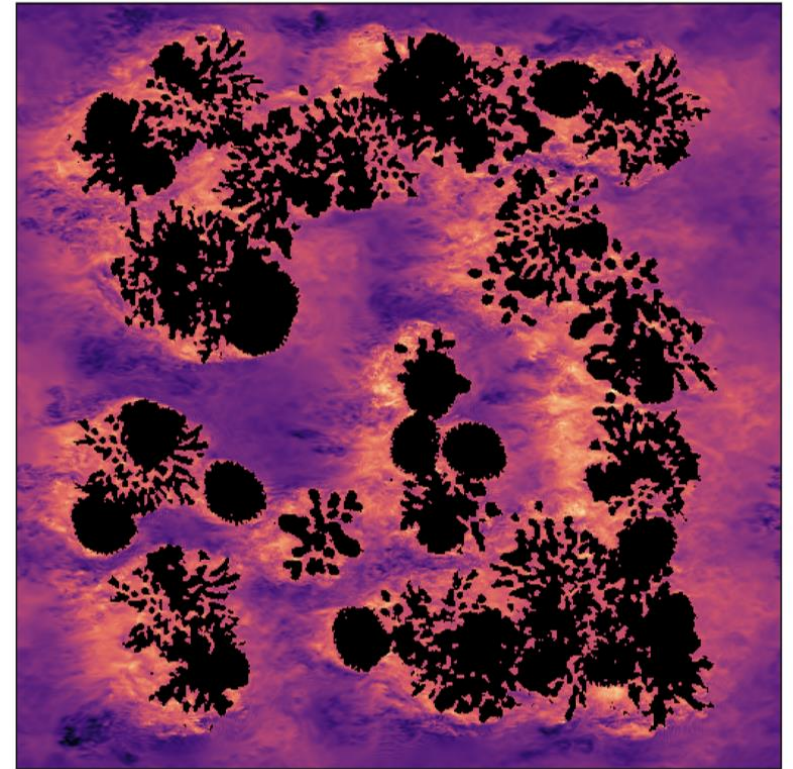


HANDOUT + CHALK BOARD

SCALES OF TURBULENT MOTION



- ☐ Governing Equations
- ☐ Scales of Turbulent Motion
- ☐ Turbulence Modelling
- ☐ Wall modelling



Turbulent Flow over a fictitious Coral Reef

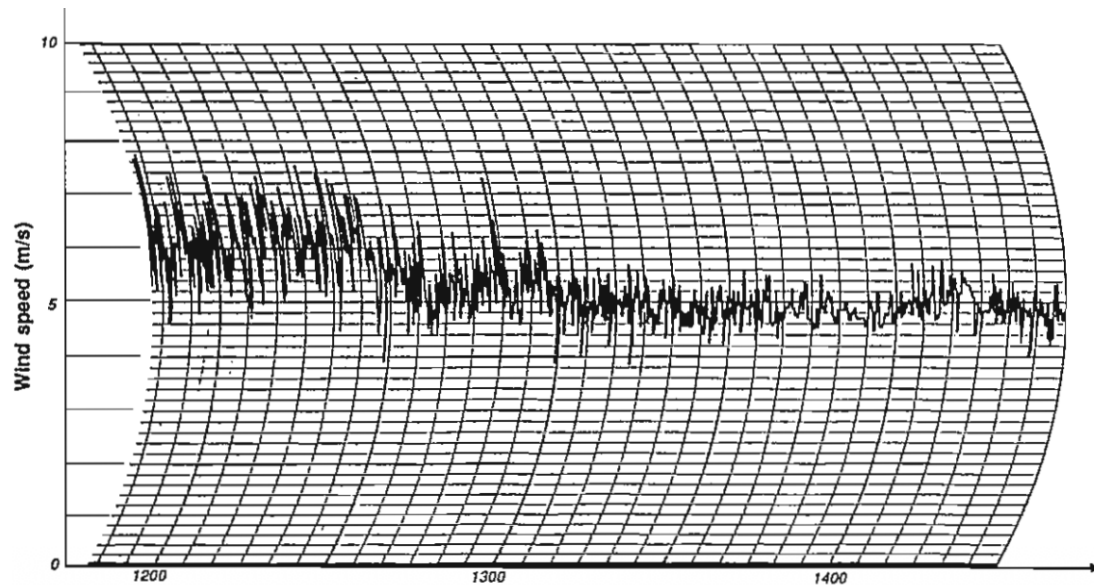
SCALES OF TURBULENT MOTION



Because turbulence is an intrinsic part of the ABL, we need to better understand it.

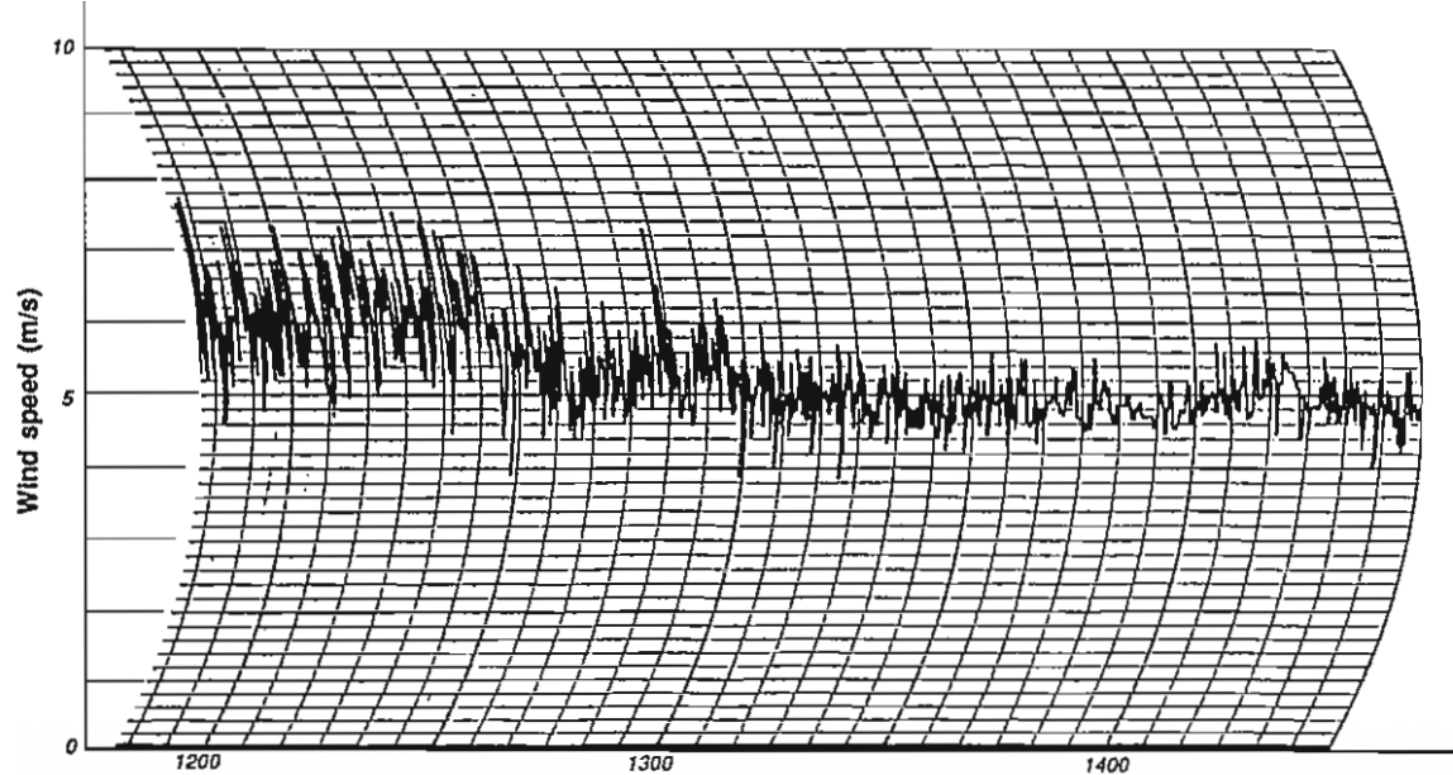
- The last unsolved classical Physics problem
- Admits, chaotic, random solutions

The randomness makes a deterministic description of turbulence difficult/impossible!



The ABL is intrinsically turbulent, look at a signal of wind speed near-surface for 2.5h

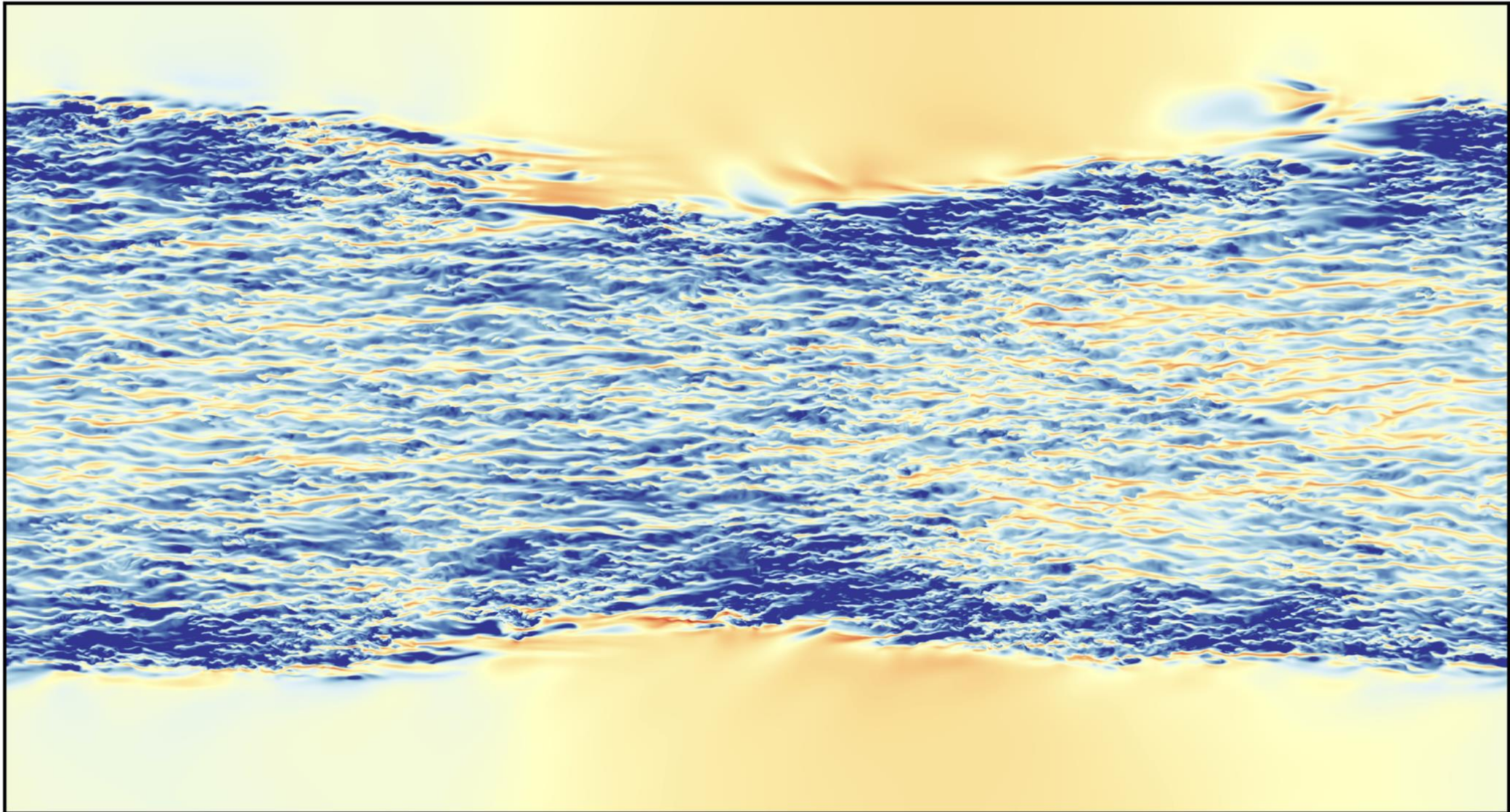
SCALES OF TURBULENT MOTION



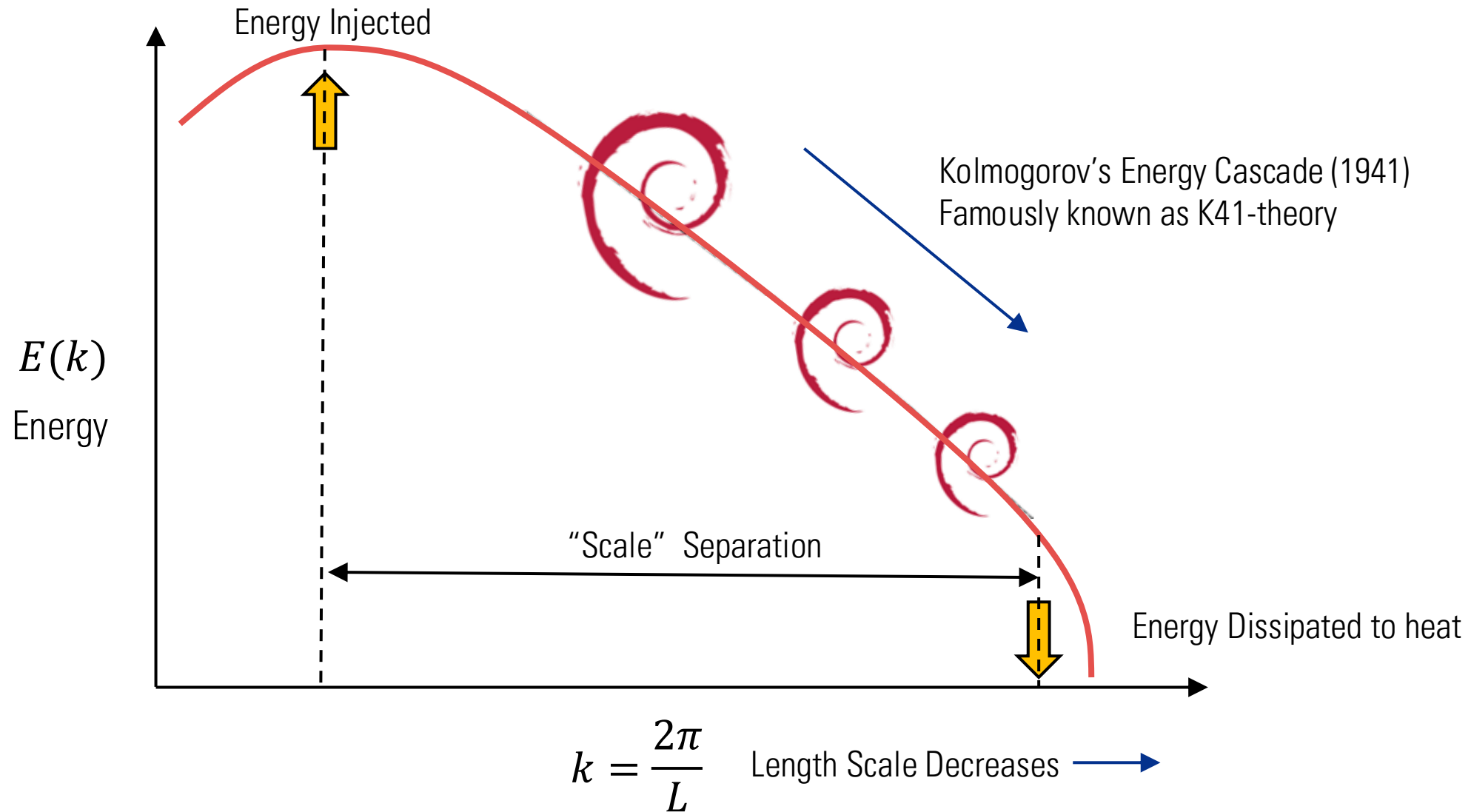
Wind speed changes irregularly, randomly, which differentiates turbulence flow from waves

There is a limited range of speed/velocity magnitude with a min and max value.

SCALES OF TURBULENT MOTION



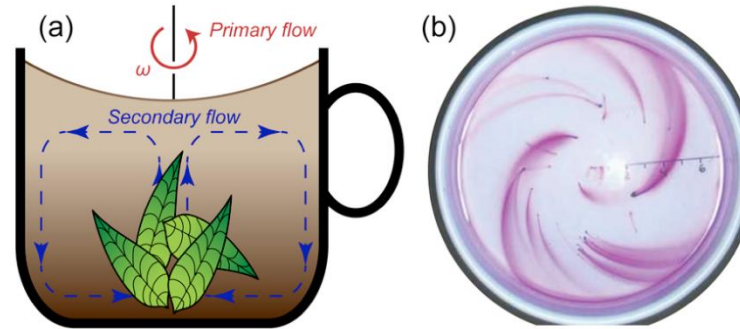
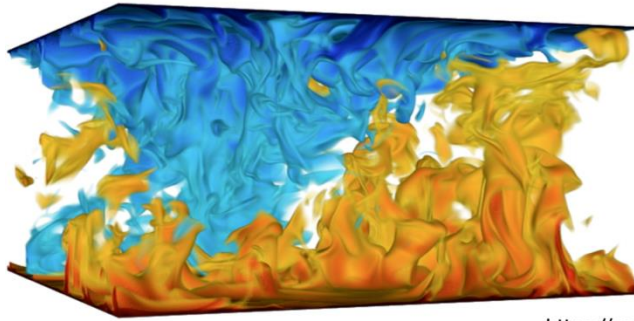
SCALES OF TURBULENT MOTION



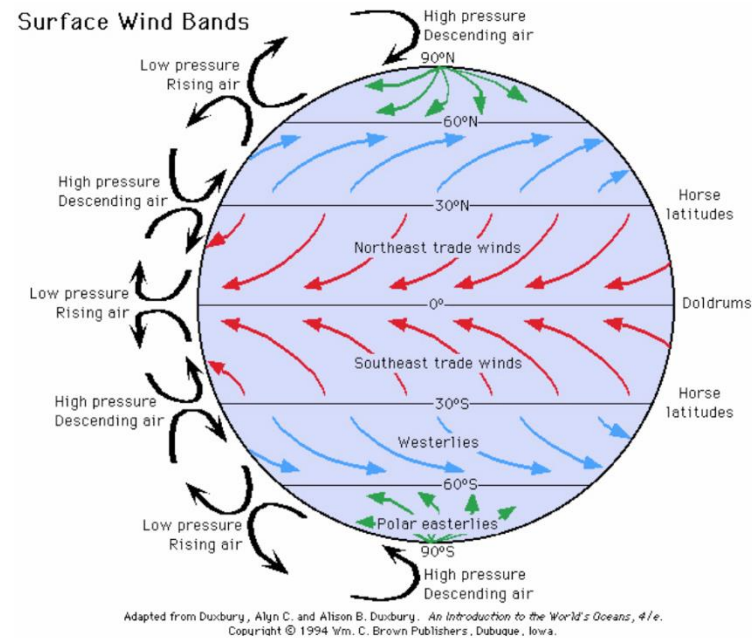
SCALES OF TURBULENT MOTION



Image(s) courtesy of Erwin P. van der Poel & Rodolfo Ostilla Mónico



https://www.researchgate.net/publication/358232909_Culinary_fluid_mechanics_and_other_currents_in_food_science

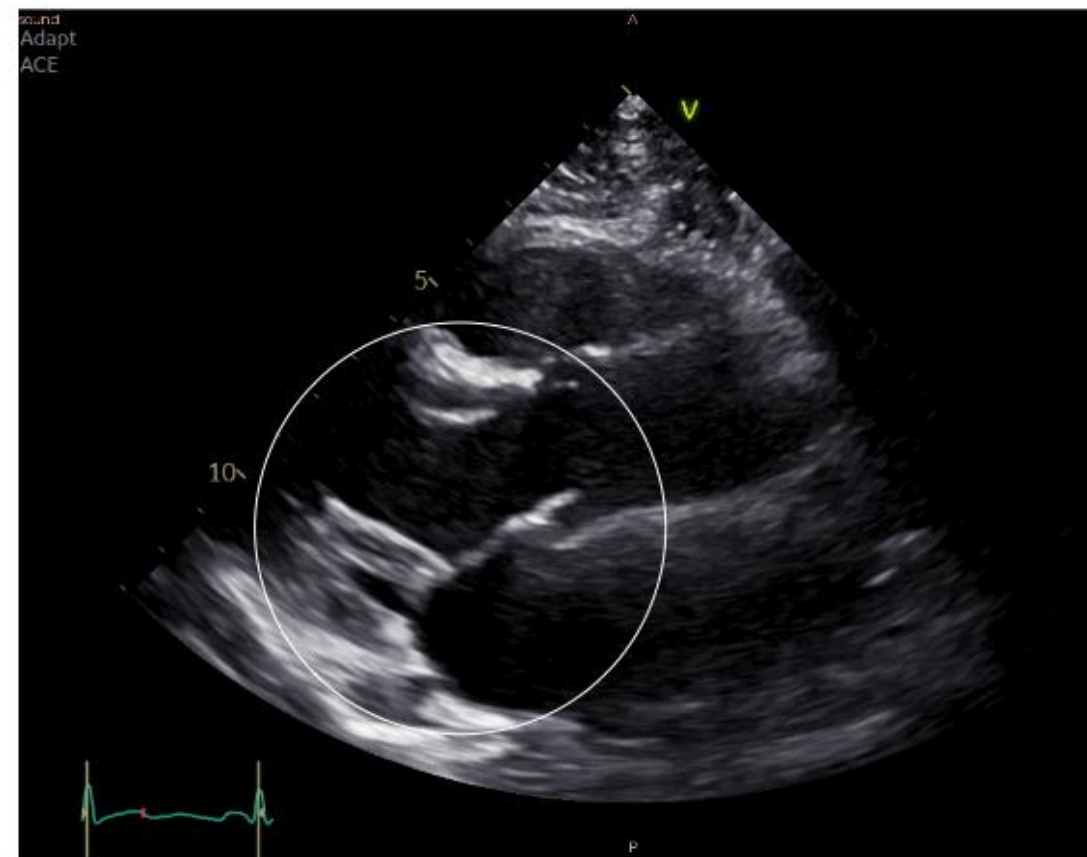
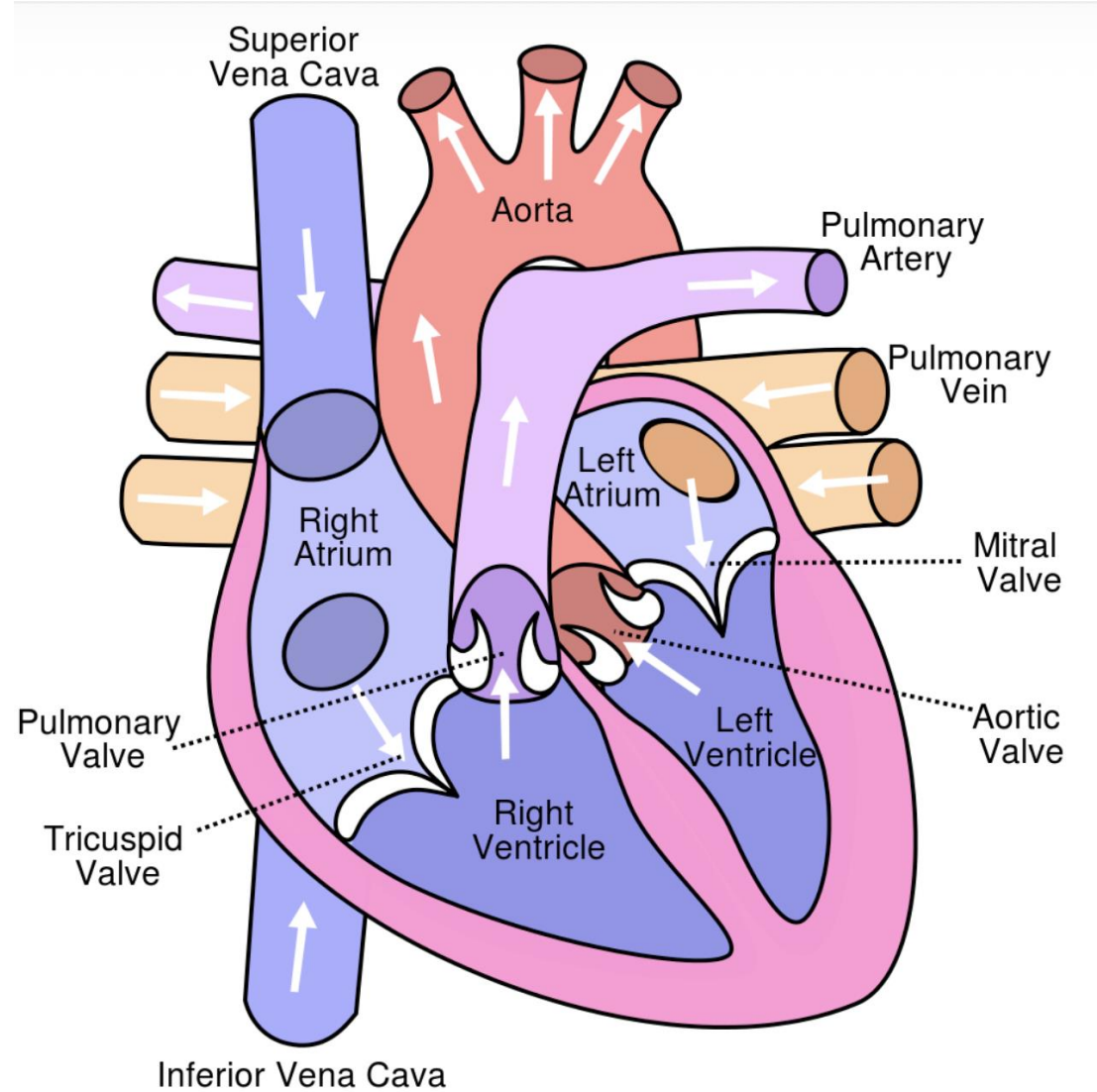


Adapted from Duxbury, Allyn C. and Alison B. Duxbury. *An Introduction to the World's Oceans*, 4/e. Copyright © 1994 Wm. C. Brown Publishers, Dubuque, Iowa.

<https://addeyans-geography.weebly.com/global-atmospheric-circulation.html>

- Multi-scale & Multi-physics
- Inherently Chaotic
- Large scale separation

SCALES OF TURBULENT MOTION



Reynolds Decomposition



Scale-decomposition

1. Separate turbulent- and mean-flow components
2. "De-couple" mean motion from turbulent motion

N = Number of velocity samples

Requirements

- a. Finite time signal – required to calculate the time-averaged velocity
- b. Isolate turbulent component from the full signal

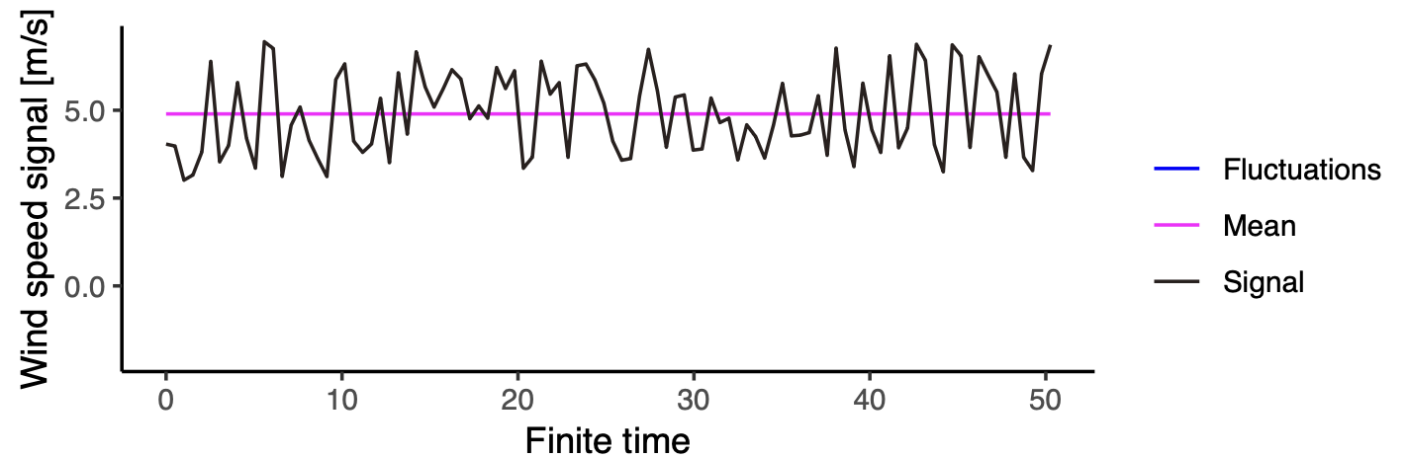
$$\bar{U} = \frac{1}{N} \sum_{t=0}^{N-1} U_t \quad u' = U - \bar{U}$$

THINK

Quadratic non-linearity in the governing equations.

Will come back to bite us a bit!

$$\partial_t u_i + \partial_j u_j u_i = g_i + \frac{1}{\rho} \partial_j \tau_{ij}$$



$$U = \bar{U} + u' \quad V = \bar{V} + v' \quad W = \bar{W} + w' \quad \theta_v = \bar{\theta}_v + \theta'_v \quad c = \bar{c} + c'$$

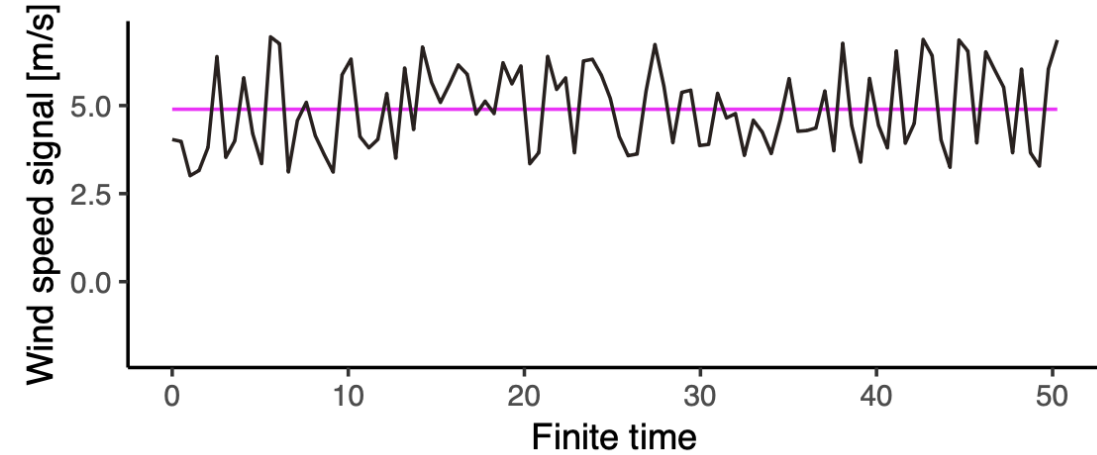
Rules for Reynolds Decomposition



Consider two time varying signals $A(t)$ & $B(t)$
and c is a scalar time invariant constant

$$\overline{(A + B)} = \bar{A} + \bar{B} \quad \overline{cA} = c\bar{A} \quad \bar{\bar{A}} = \bar{A}$$

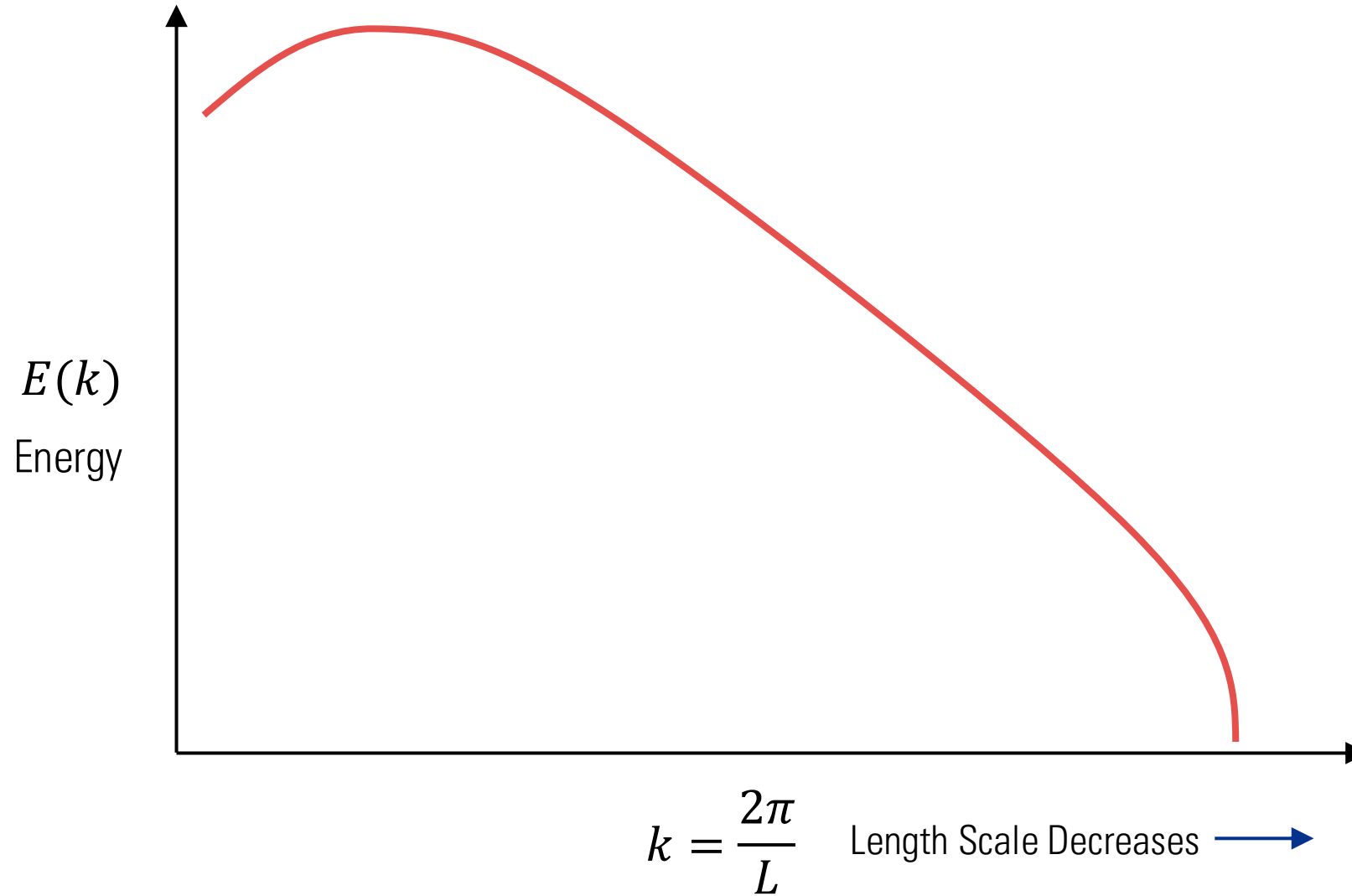
$$\overline{\bar{A}\bar{B}} = \bar{A}\bar{B} \quad \overline{\frac{dA}{dt}} = \frac{d\bar{A}}{dt}$$



Reynolds-Averaged Navier-Stokes Equations Derivation

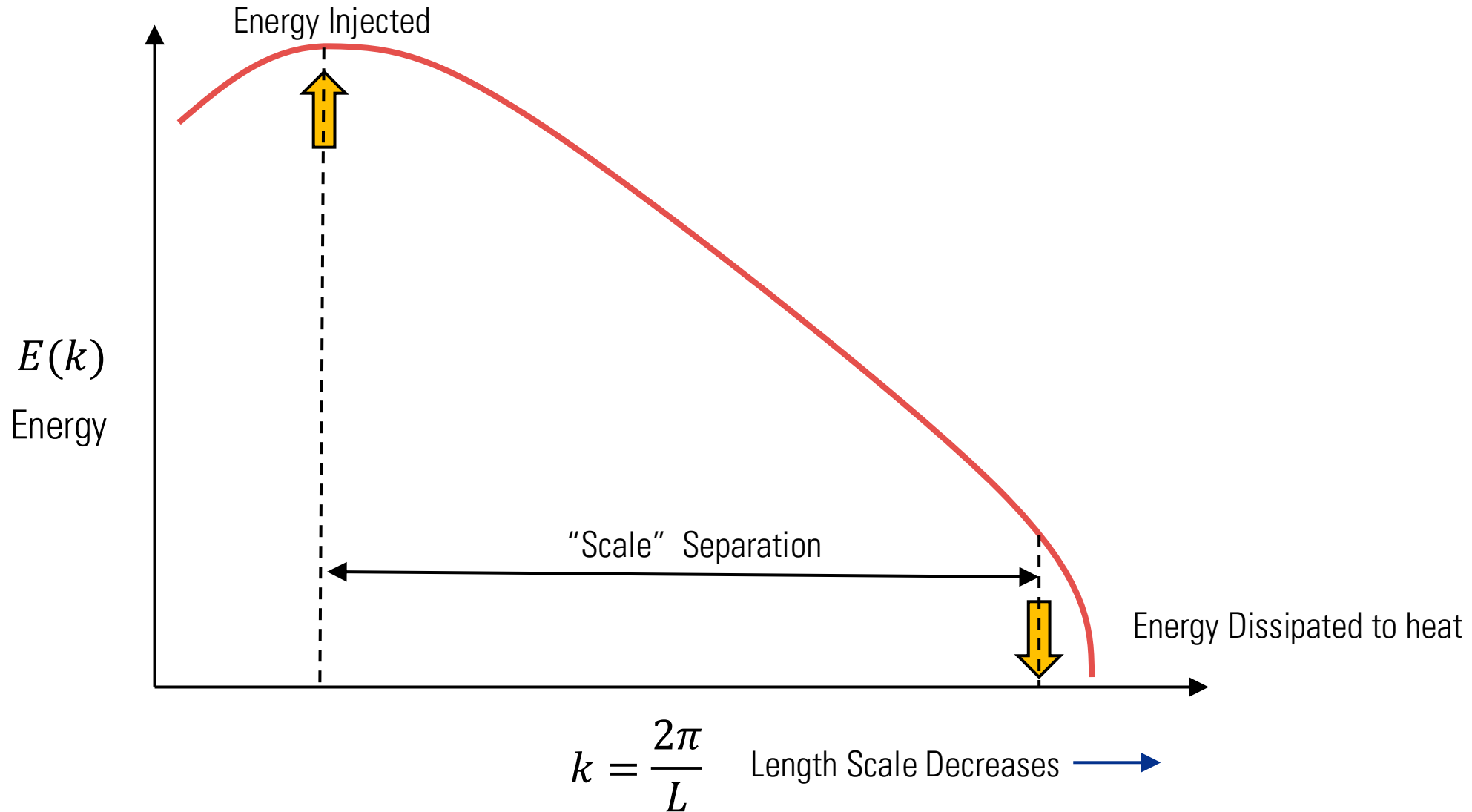
TURBULENCE MODELLING

The “art” of drawing boundaries!



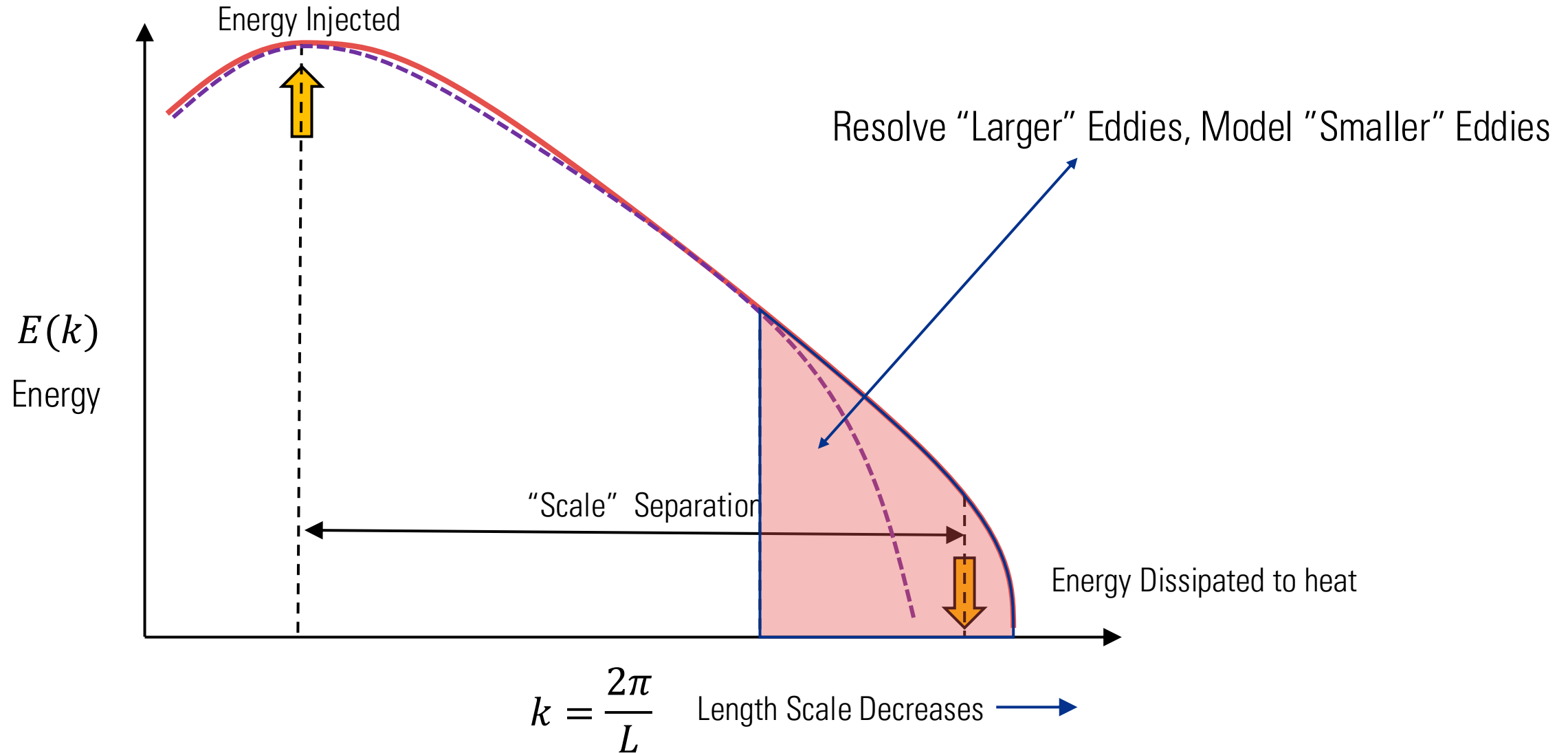
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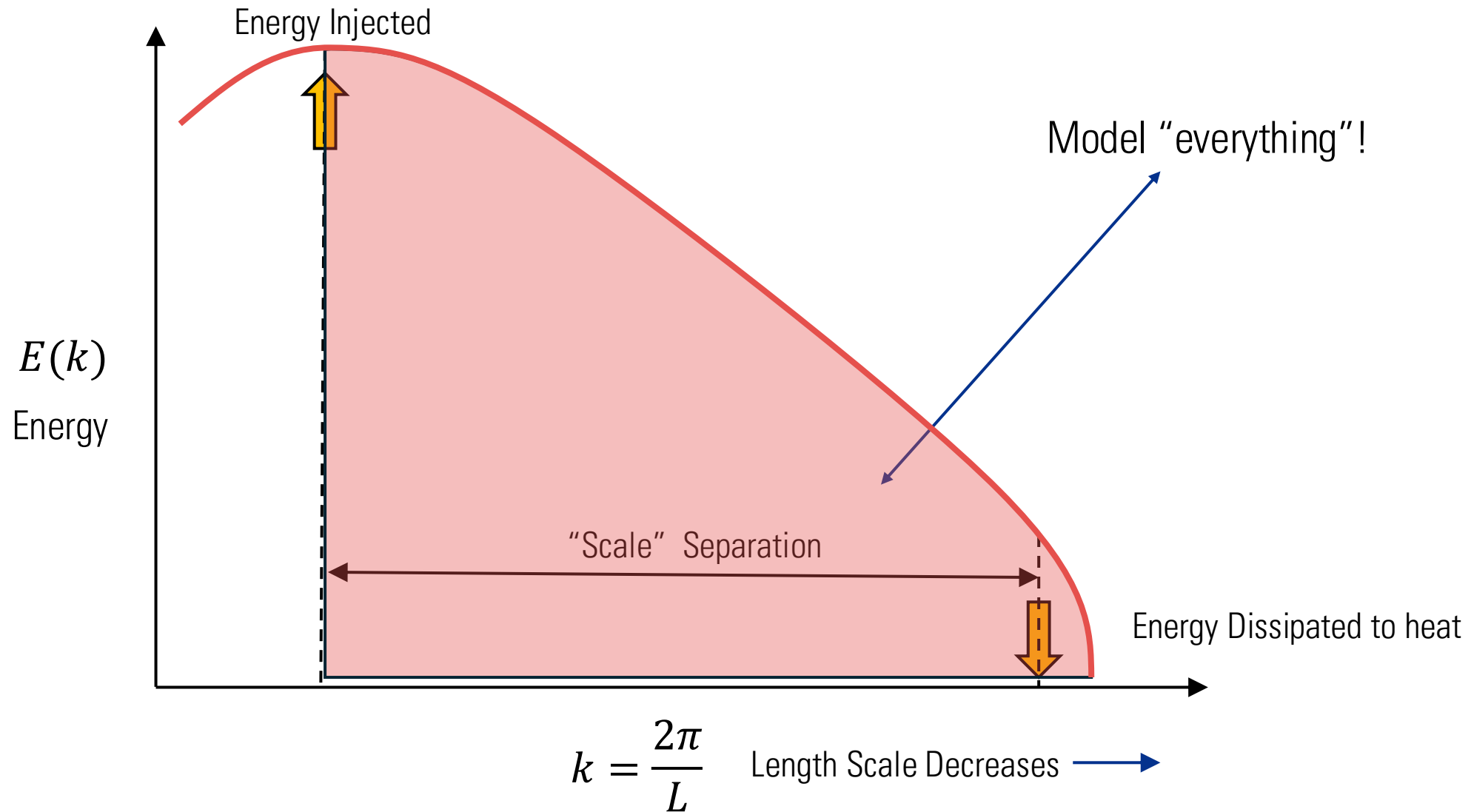
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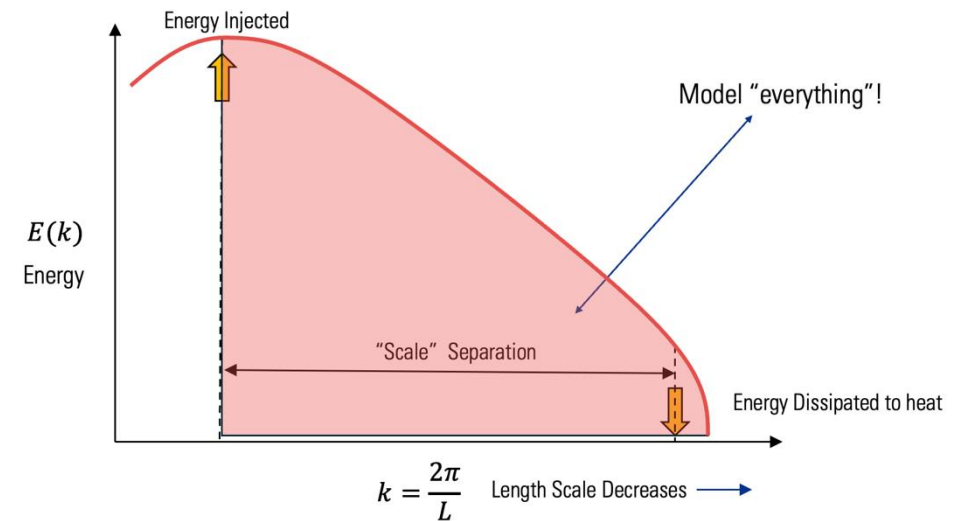
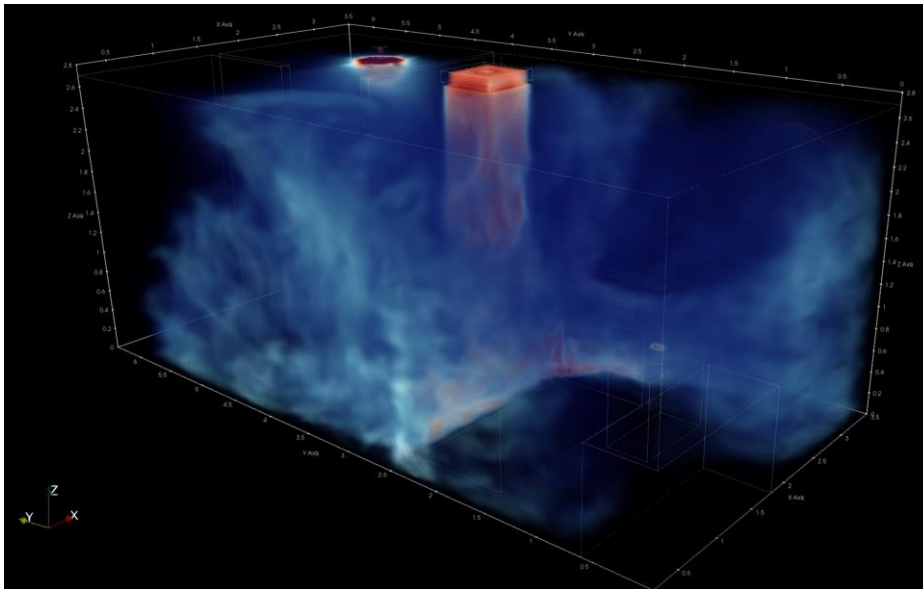


$$\partial_j k \bar{u}_i = \partial_j \left(\nu + \frac{\nu_t}{\sigma_k} \partial_j k \right) + P_k - \epsilon + S_k$$

Transport of TKE

$$\partial_j \epsilon \bar{u}_i = \partial_j \left(\nu + \frac{\nu_t}{\sigma_\epsilon} \partial_j \epsilon \right) + C_{1,\epsilon} S_\epsilon - C_{2,\epsilon} \frac{\epsilon^2}{k + \sqrt{\nu \epsilon}} + S_\epsilon$$

Transport of Dissipation



Video Credit: A. Giri: Indoor ventilation LES

WALL MODELLING

- Momentum Mixing: How to model the physics when the numerical fidelity cannot resolve the wall physics?
- We use established "Equilibrium" physics to guide the decision making

